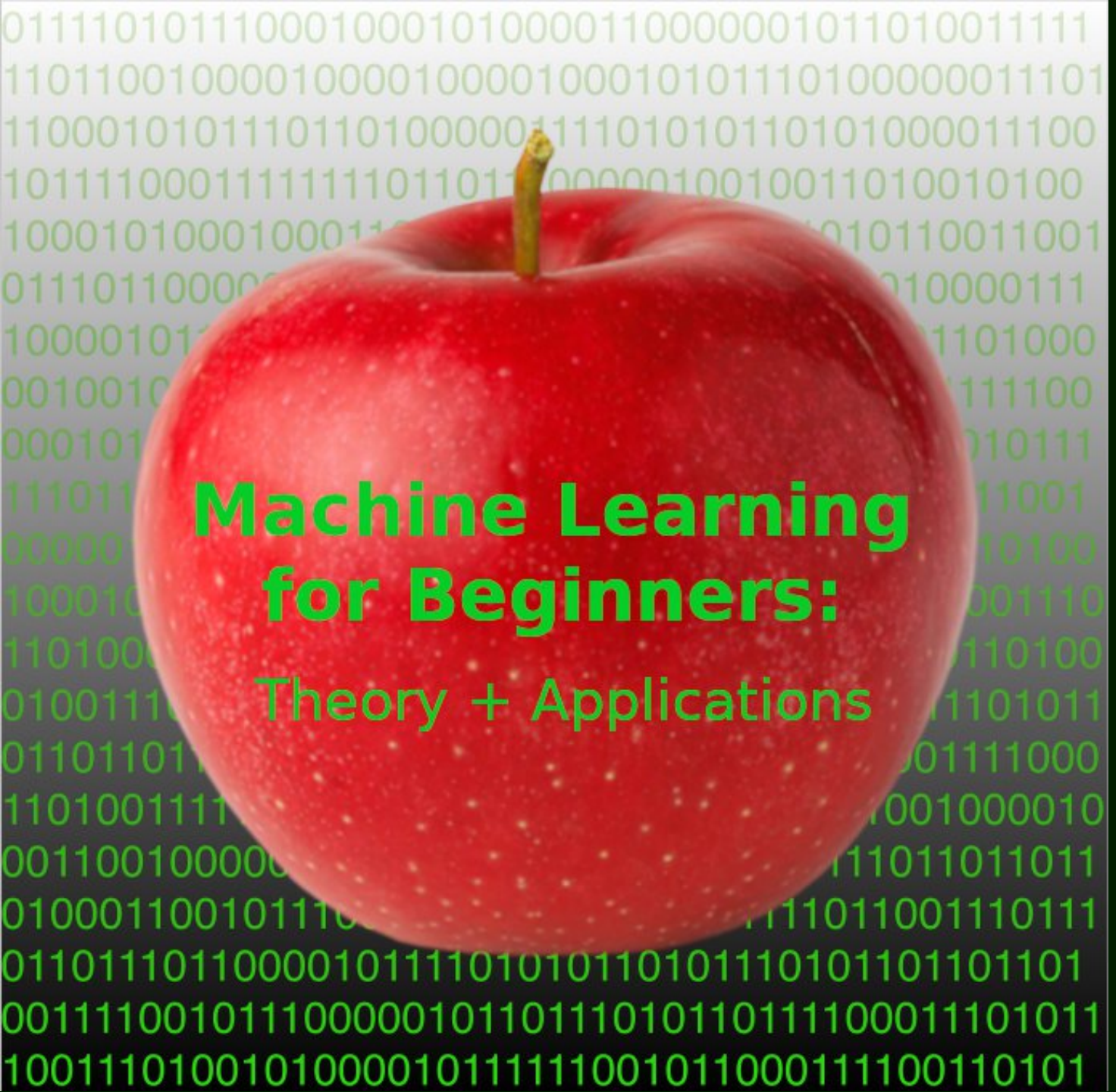


University of Bayreuth
Chair for Polymer Engineering
Prof. Dr. Holger Ruckdäschel

Title:

"Machine Learning for Beginners: Theory and Applications"

(Available in E-learning/Moodle)

Lecturer:

Dr. Rodrigo Q. Albuquerque

Lecture 3

Topics

Week 1: Introduction
ML types, data, ethics, case studies

Week 2: Preprocessing & *k*NN
Preprocessing, *k*-nearest neighbors, distance metrics, hyperparameters, cross-validation

★ **Week 3: Linear Regression**
Types, loss function, (over/under)fitting bias/variance, L1/L2 regularization, metrics

Week 4: Classification
Logistic regression, SVM, classification evaluation metrics, confusion matrix one-hot/label encoding

Week 5: Random forests
Decision trees, random forest, classification and regression applications

Week 6: PCA / KMeans
Dimensionality reduction, clustering

Week 7: Extra exercises

Week 8: Simple ANNs
Feedforward networks, activation functions, basic training

Week 9: Gaussian Processes
Kernels, gaussian processes

Week 10: Design of Experiments
DoE, random design, active learning, bayesian optimization

Weeks 11-12: Student projects

Linear regression

Simple linear regression

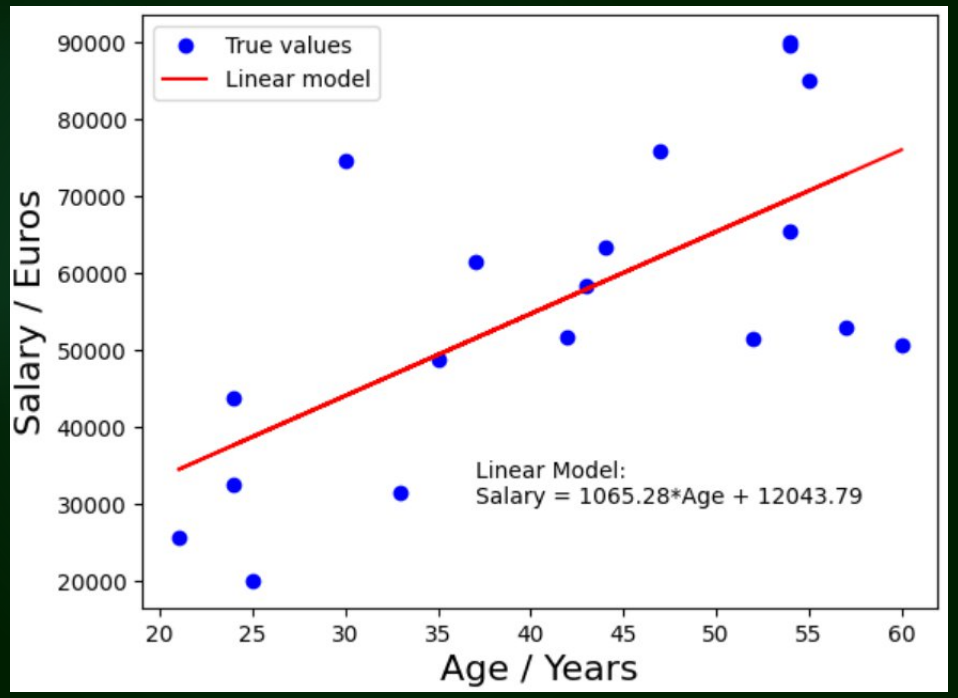
Model: $\hat{y}_i = w_0 + w_1 x_i$

bias (or intercept)

1 feature (= Age)

weight (or slope)

Predicted target for sample *i*



Data: “employees_clean_encoded.csv”

Regression task: Find w's given *n* data points (x_i, y_i)

Method: Minimize the loss function ($\mathscr{L}$)

$$\mathscr{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$ (= Mean squared error, MSE)

True (or expected) property

$$\frac{\partial \mathscr{L}}{\partial w} = 0$$

Matrix solution: $w_{LS} = \left(X^T X\right)^{-1} X^T y$ $\rightarrow$ $w_{LS} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$

New predictions (matrix form): $\hat{y}_{new} = X_{new} w_{LS}$ (X_{new} = samples as rows)

Ex: if Age = 50, predicted salary = 50 * 1065.28 + 12043.79 = 65307.79

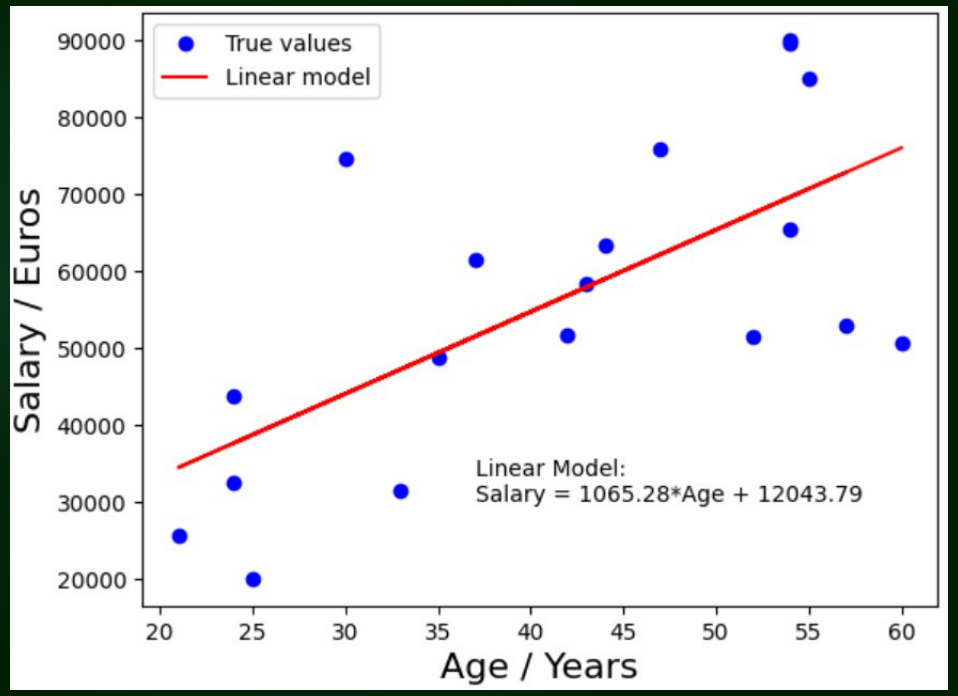
In practice: we use the sklearn library in Python!

Linear regression

Simple linear regression

Trained model: $\hat{y}_i = 12043.79 + 1065.28x_i$

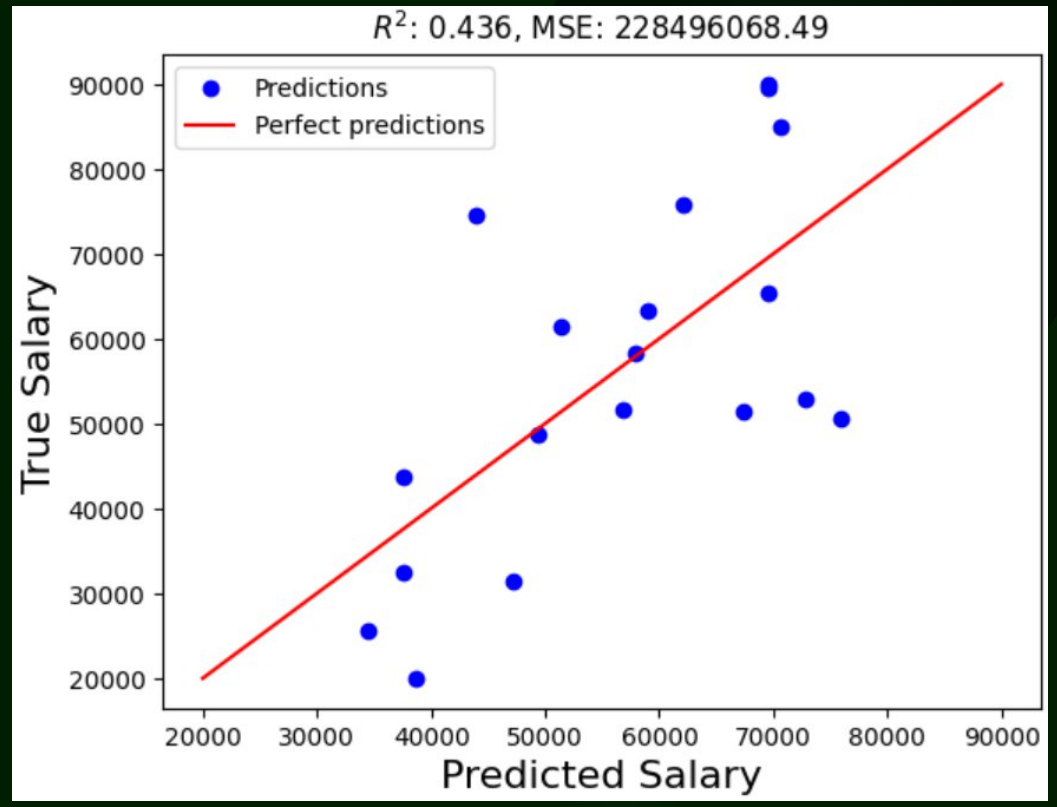
Model visualization (possible for low dimensions only!)



Data: “employees_clean_encoded.csv”

PARITY PLOT

Visualization of model performance (ALWAYS possible)



Regression tasks: ALWAYS show parity plots!!

Linear regression

Multiple linear regression

$$\widehat{salary} = w_0 + w_1 Age + w_2 Experience$$

2 weights!

Model: $\hat{y}_i = w_0 + w_1 x_{i,1} + w_2 x_{i,2}$

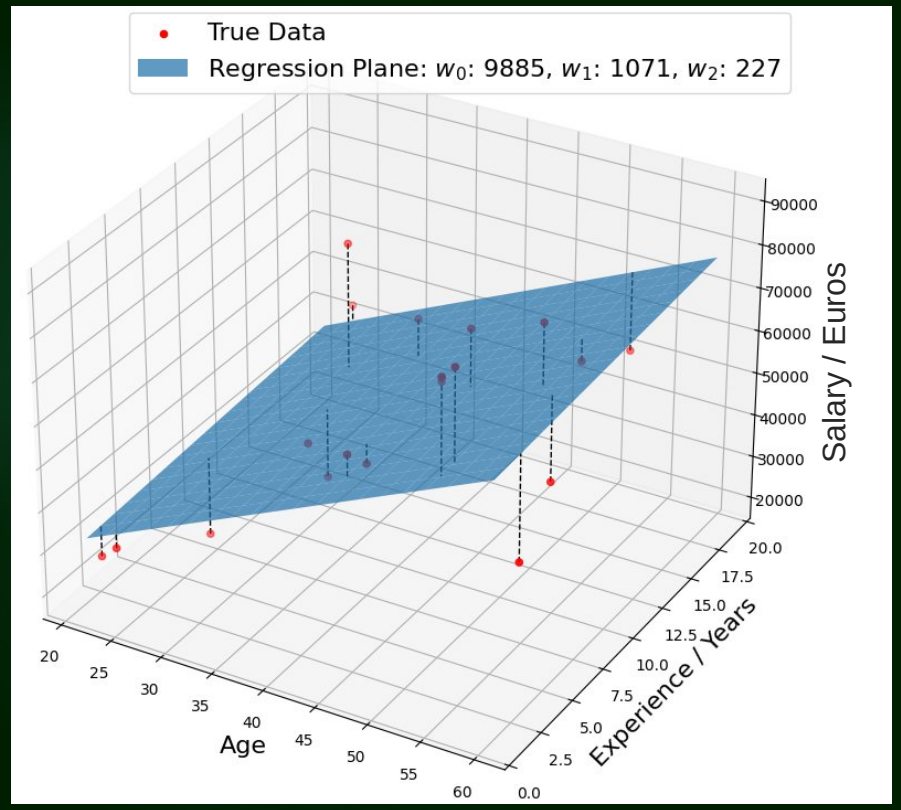
bias

2 features: Age, Experience

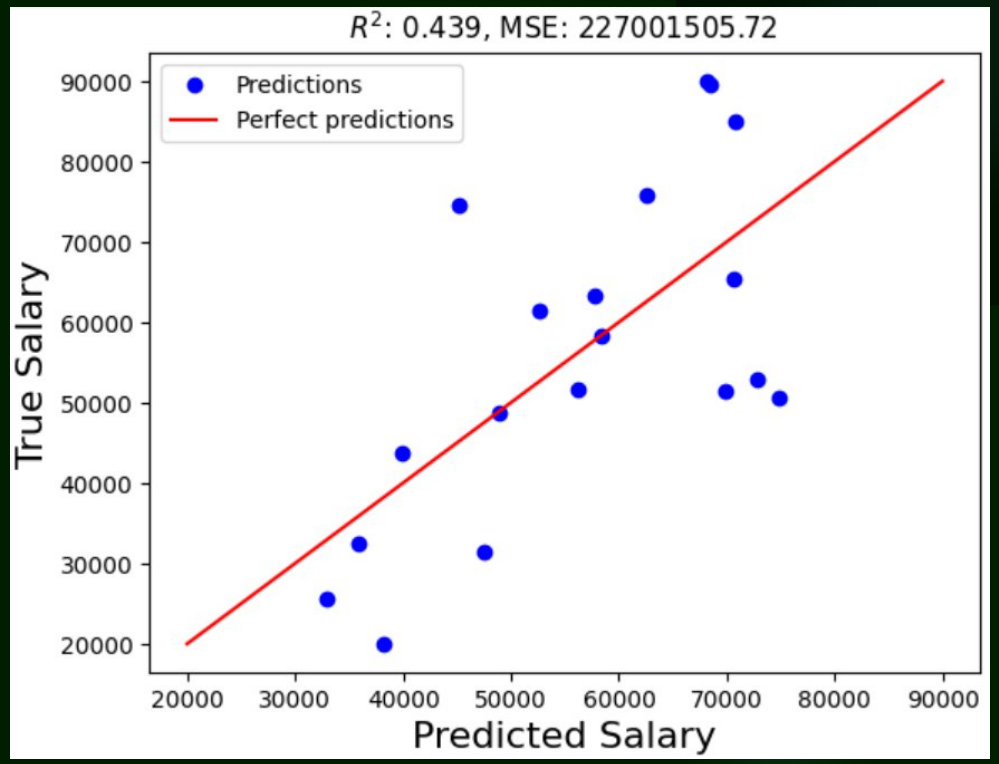
Matrix solution: $w_{LS} = \left(X^T X\right)^{-1} X^T y$ → $w_{LS} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$

New predictions:

Ex: if Age = 50 and Experience = 10: **Pred. salary:** 9885 + 1071*50 + 227*10 = 65705



Data: “employees_clean_encoded.csv”

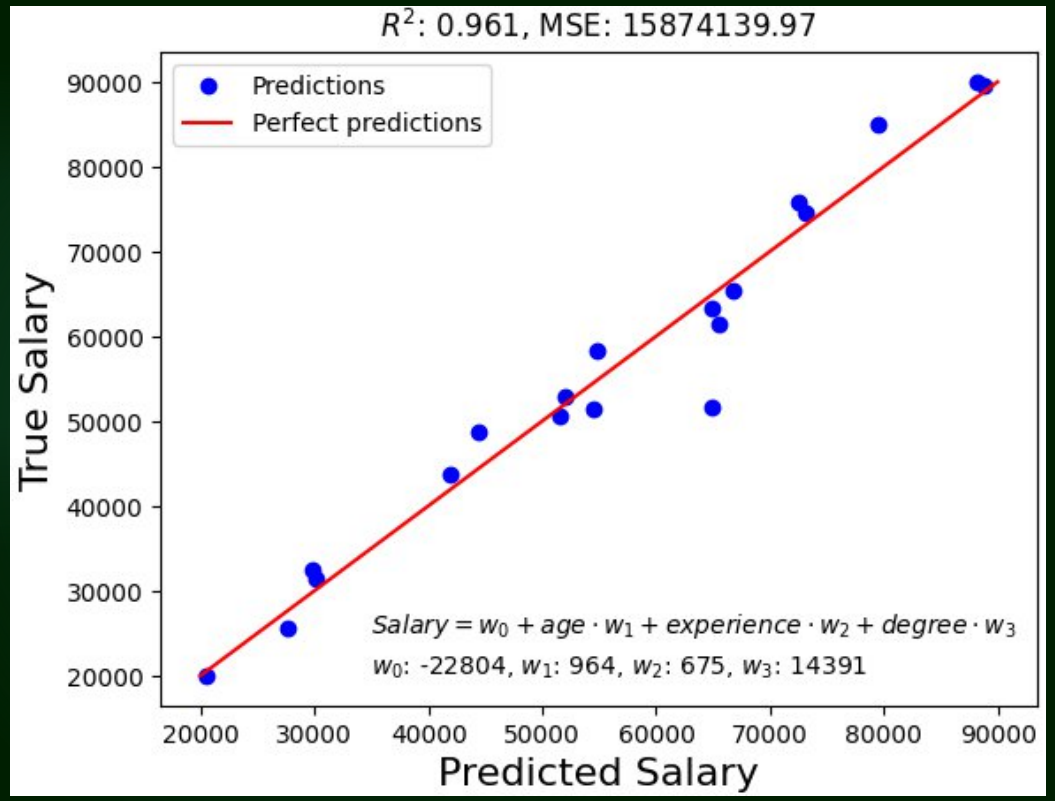


Parity Plot

Linear regression

Multiple linear regression

$$\widehat{salary} = w_0 + w_1 Age + w_2 Experience + w_3 HighestDegree$$



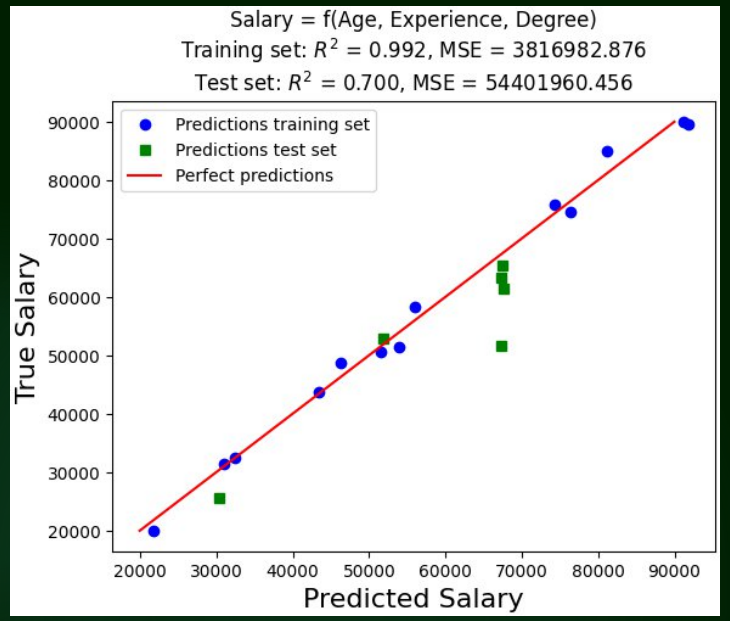
Data: “employees_clean_encoded.csv”

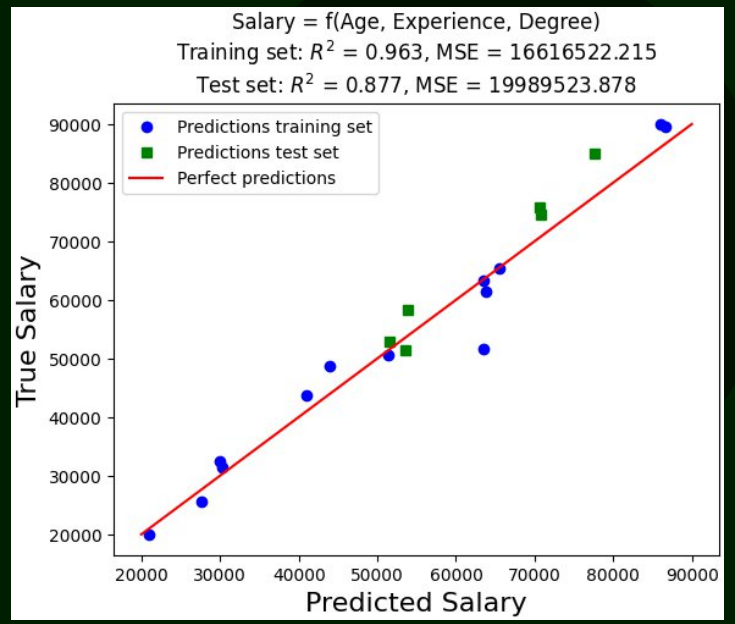
PARITY PLOT

Up to now, no dataset splitting was used!
Let’s split the dataset into training and test sets

Linear regression

Random splittings (test set: 30%)

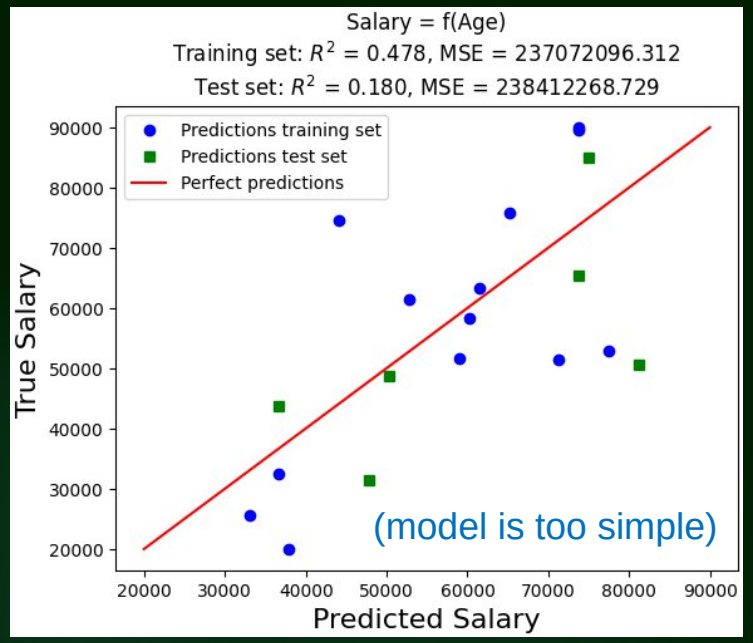


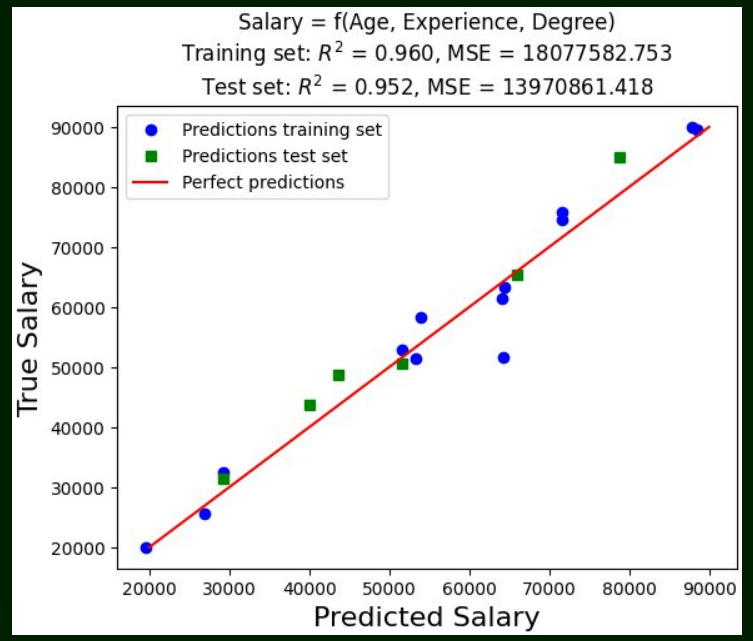


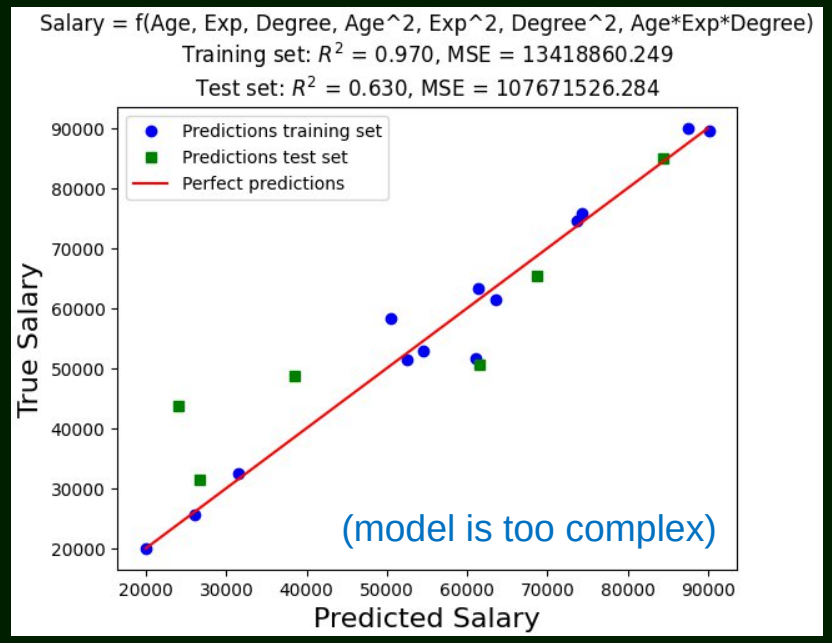
Data: “employees_clean_encoded.csv”

- Training errors (MSE_{train}) < test errors (MSE_{test})
- The model performance fluctuates a lot depending on the data used to train it (high model variance)

Overfitting x underfitting







Data: “employees_clean_encoded.csv”

$MSE_{train,test}$ ⬆ (~ 10^8)

$MSE_{train,test}$ (~ 10^7)

MSE_{train} (~ 10^7) << MSE_{test} (~ 10^8)

Underfitting ← - model complexity + → Overfitting

The model cannot predict at all!
It does not capture any pattern in the data

The model is poor on the test set
Poor generalization
Model too specialized in the training set

Regularization

Note: **ElasticNET** combines L1 and L2

Lasso regression (L1 norm)

$$\mathscr{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\sum_{j=1}^{m}|w_j|$$

Upon increasing λ , the L1 norm shrinks w's of less important features, many of them becoming zero (= feature selection):

- Model interpretability is easier
- Faster training for large data
- Better to treat sparse data

w_{LR} = solved iteratively (no closed form)

Ridge regression (L2 norm)

$$\mathscr{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\sum_{j=1}^{m}w_j^2$$

Upon increasing λ , the L2 norm shrinks w's of less important features, but they will never become zero:

- Model interpretability is more difficult
- Slower training for large data
- "*Kernel trick*" is possible

$$w_{RR} = \left(\lambda I_n + X^T X\right)^{-1} X^T y$$

λ is a hyperparameter!

Regularization

Ridge regression

(employees.csv dataset)

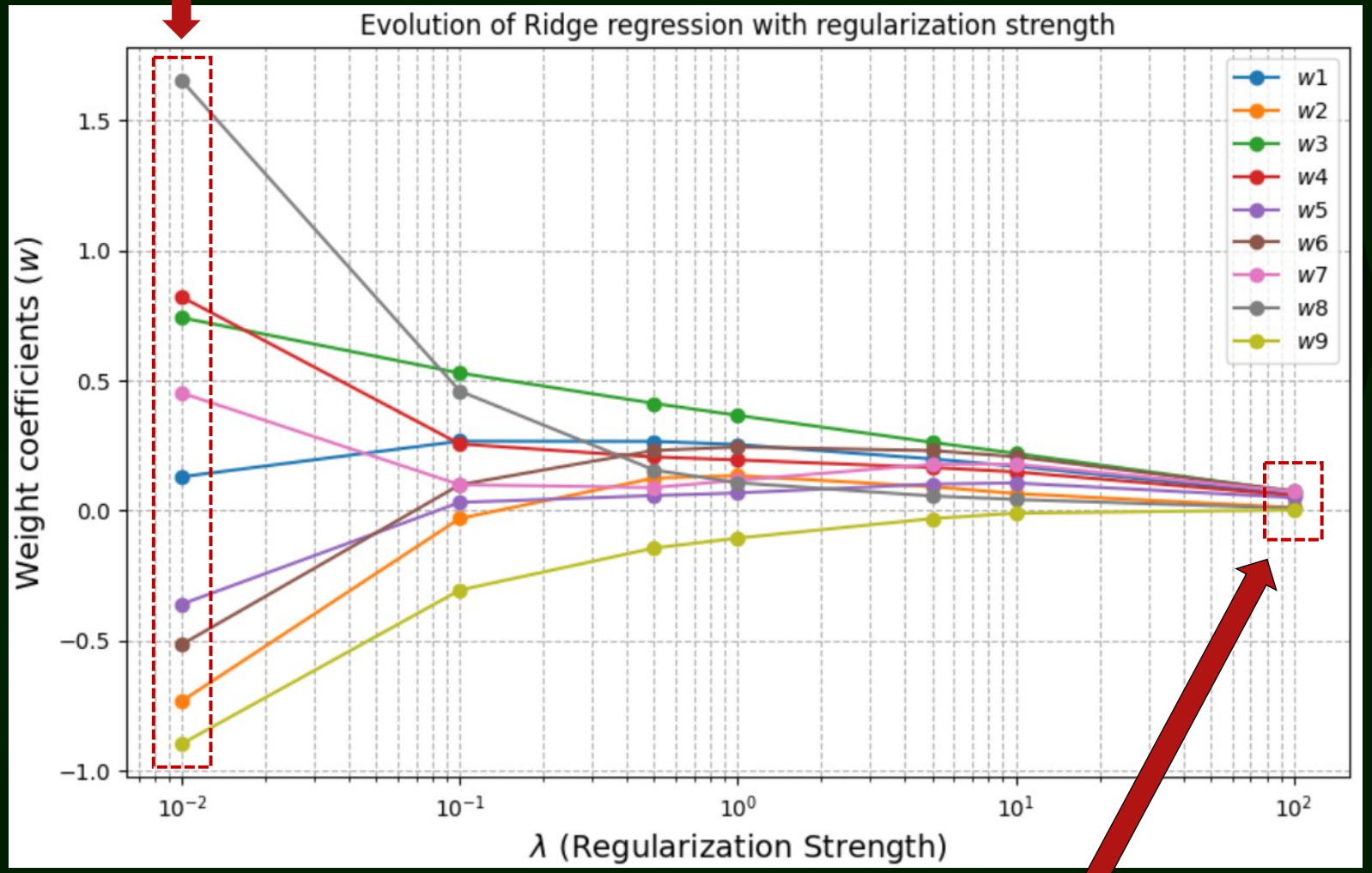
$$\mathscr{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\sum_{j=1}^{m} w_j^2$$

$$\hat{y}_i = w_1 f_1 + w_2 f_2 + \cdots + w_9 f_9$$

What is the best λ?

– Lowest Error (CV)

$$\hat{y}_i = 0.13 f_1 - 0.73 f_2 + \cdots - 0.89 f_9$$



$$\hat{y}_i = 0.062 f_1 - 0.011 f_2 + \cdots - 0.001 f_9$$

Exercises

1) Download the file ‘files3.zip’ from moodle/e-learning (folder “Week 3”) and unzip it locally on your computer
2) Go to google colab (online) **or** launch jupyter notebook locally (offline)
3) Open/upload ‘exercise3.ipynb’ and wait for more instructions